Definition of OR -

Operations research is the systematic application of quantitative methods, techniques and tools to the analysis of problems involving the operation of systems

--> Operations research as a technique for selecting the best course of action out of the several courses of action available, in order to reach the desirable solution of the problem.

Features of OR -

1. Technically Appropriate: The solution should work technically, meet the constraints and operate in the problem environment.
2. Reliable: The solution must be useful for a reasonable period of time under the conditions for which it was designed.
3. Economically Viable: Its economic value should be more than what it costs to develop it and it should be seen as a wise investment in hiring OR talent.
4. Behaviourally Appropriate: The solution should be behaviourally appropriate and must remain valid for a reasonable period of time within the organization.

Applications of OR -

1. Logistics and Supply Chain Management - Route Optimization, Inventory Management, Warehouse Optimization
3. Manufacturing - Production Planning and Scheduling, Resource Allocation, Quality Control Facility Layout
4. Transportation - Airline Scheduling, Traffic Flow Management, Public Transportation Planning, Vehicle Routing
5. Finance - Portfolio Optimization, Risk Management, Financial Forecasting, Credit Risk Analysis
6. Healthcare - Hospital Resource Allocation, Patient Scheduling, Emergency Services Optimization Disease Modeling
7. Telecommunications - Network Design, Capacity Planning, Call Center Optimization
8. Energy - Power Grid Optimization, Renewable Energy Integration, Energy Demand Forecasting
9. Public Sector - Urban Planning, Emergency Response Planning, Resource Allocation

Opportunities in OR -

- decision-maker to define explicitly his/her objectives, assumptions and constraints.
- decision-maker to identify variables that might influence decisions.
- in identifying gaps in the data required to support solution to a problem.
- the use of computer software to solve a problem

Limitations of OR -

- The solution to a problem is often derived by simplifying assumptions. Consequently, such solutions are not optimal.
- Models may not represent the realistic situations in which decisions are made.
- Often the decision-maker is not fully aware of the limitations of the models.

Methodology of OR

Step 1: Defining the problem -

Clearly identify and define the problem.
Determine the objectives and constraints.
Establish the scope of the study.

Step 2: Formulating a mathematical model -

Develop a mathematical or logical model that represents the problem.
Identify decision variables, objective function, and constraints.
Choose an appropriate type of model (e.g., linear programming, network model, simulation).

Step 3: Solving the mathematical model-

Select an appropriate solution technique (e.g., simplex method, network algorithms, simulation).
Use computer software to solve the model.
Validate the solution.

Step 4: Validating the solution -

Test the model to ensure it accurately represents the real-world problem.
Validate the model's assumptions and results.
Perform sensitivity analysis to assess the impact of changes in input parameters.

Step 5: Implementing the solution -

Implement the solution in the real-world system.
Monitor the performance of the implemented solution.
Make adjustments as needed

Step 6: Modifying the model-

Step 7: Establishing controls over the solution -

A control procedure needs to be developed for detecting significant changes in decision variables of the problem, without having to build a model every time a significant change occurs.

Advantages of OR

1. **Structured approach to problems:** A substantial amount of time and effort can be saved in developing and solving OR models if a logical and consistent approach is followed. This implies that the decision maker has to be careful while defining variables, availability of resources, and functional relationships among variables in the objective function and constraints. This will also reduce the chance of conceptual and computational errors. Any such error can also be detected easily and corrected at an early stage.

2**. Critical approach to problem solving:** The decision-maker will come to understand various components of the problem and accordingly select a mathematical model for solving the given problem. He will become aware of the explicit and implicit assumptions and limitations of such models. Problem solutions are examined critically and the effect of any change and error in the problem data can be studied through sensitivity analysis techniques

OR models on use

Linear Programming (LP)
Integer Programming (IP)
Network Models
Queuing Models
Simulation Models
Decision Analysis
Game Theory

Computer Software for OR

Quantitative Systems for Business Plus (QSB+), Version 3.0
Quantitative Systems for Operations Management (QSOM)
Value STORM: MS Quantitative Modelling for Decision Support
Excel 97
Linear Interactive Discrete Optimization (LINDO)